Edge-Optimized Voice Control with 0.26M Parameters: Distilling 86M Adaptive Window Audio Transformer for Real-World Variable-Length Inputs

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***Abstract*—Intelligent voice control devices in edge computing require lightweight models capable of processing real-world variable-length audio inputs. While transformer-based models (e.g., AST, BEATs) achieve high accuracy, their fixed-length input constraints and excessive parameters (e.g., 87M in AST) hinder deployment in resource-constrained edge computing environments. To address this, we propose KD-AWAT, a framework combining an Adaptive Window Audio Transformer (AWAT) and extreme knowledge distillation. AWAT introduces a dynamic hop-length mechanism to convert variable-duration audio (1.0–8.5s) into fixed-size Mel spectrograms (224×224), enabling compatibility with pre-trained ViT architectures while preserving temporal integrity. Subsequently, we distill AWAT's 86M parameters into a 0.26M-parameter MiniNet through dual compression: width multiplier scaling and adaptive expansion ratios, coupled with hybrid KL-divergence and cross-entropy loss. Evaluated on a custom dataset (5,000+ samples, 24 classes, variable lengths), KD-AWAT achieves 330× compression while surpassing original AWAT's accuracy (96.6% vs. 96.0%) and also outperforming AST (90.9%). Our framework resolves the conflict between edge deployment constraints and real-world variability, demonstrating practical viability in industrial voice control systems.**

Keywords*—Variable-length audio inputs, Edge computing, audio transformer, adaptive window, knowledge distillation.

I. Introduction

Voice control interfaces are rapidly becoming the backbone of human-machine interaction in edge computing ecosystems, enabling real-time command execution in applications ranging from autonomous robotics to smart industrial systems [1]-[6]. However, deploying these systems on resource-constrained devices faces two fundamental barriers: (1) the prohibitive computational costs of state-of-the-art audio models, and (2) their inability to handle variable-length voice inputs, which is a critical requirement for real-world usability.

Modern transformer-based audio classification architectures, such as Audio Spectrogram Transformer (AST) [7], have set new benchmarks in speech recognition accuracy. Yet, their practical adoption is limited by rigid design choices. For instance, one of datasets AST utilized is AudioSet [8], where all 10-second audio clips are padded or truncated to a fixed length, ignoring natural variations in voice command durations and human speech habits variance. Similarly, EAT [9], a self-supervised audio transformer model, which processes ESC-50 [10], AudioSet data requires fixed-length input. Moreover, SepTr [11], a separable transformer for audio spectrogram processing, also requires fixed-length input when constructing model. BEATs [12], an audio pre-training with acoustic tokenizers, which is tested on ESC-50 and AudioSet also needs fixed input when dealing with different tasks.

These limitations are particularly acute in high-performance computing (HPC)-edge scenarios, where voice interfaces must operate under strict energy budgets while processing unpredictable input lengths. For example, in distributed sensor networks, a single edge node might receive 1.0s diagnostic queries and 8.5s control commands within milliseconds. Current models fail in such environments: AST's 87M parameters demand excessive memory and is uncapable of dealing with mixed varied length audio commands.

To address these challenges, we propose KD-AWAT (Knowledge-Distilled Adaptive Window Audio Transformer), a framework that bridges transformer-level accuracy with edge deployment practicality through two innovations:

1. **Native Variable-Length Support:** Adaptive Window Audio Transformer (AWAT) utilizes an adaptive window mechanism that eliminates AST's fixed-length constraints, enabling robust handling of variable-duration voice commands (1.0s–8.5s) with significantly accuracy than AST in real-world tests.
2. **Extreme Model Compression:** By distilling the 86M-parameters AWAT into a 0.26M-parameters student model, we achieve 330× parameter reduction while surpassing teacher's accuracy, revealing a great potential on energy-bounded edge device implementation.

We validate KD-AWAT on a custom dataset of over 5,000 real-world voice commands with 24 different classes and variable length. Moreover, we tested pretrained AST model

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on the same dataset by reshaping the spectrograms obtained without our adaptive window into same sizes. Results show that our 0.26M model significantly outperforms AST, making it applaudable for real edge devices utilization.

The rest of the paper is organized as follows: Sections II further introduces the related research work of our work. In Section III, we describe our research methods. The effectiveness of our approach is evaluated on our customed varied length dataset in Section IV. Finally, Section V summarizes this paper.

II. Background and Related Work

A. Audio Classification

Audio classification is a core subtask within Automatic Speech Recognition (ASR), enabling computers to interpret audio input and transcribe speech into text. Traditional ASR systems leverage spectrograms as an intermediate representation, optimizing feature extraction via Convolutional Neural Networks (CNNs) for low-latency processing. While recurrent neural networks (RNNs) like Bi-LSTM [12] have dominated ASR since their introduction by Graves in 2013 (particularly in IoT devices like smart speakers), recent advances integrate CNNs with RNNs and Deep Neural Networks (DNNs). Architectures such as CLDNN [13] exemplify this hybrid approach, achieving state-of-the-art accuracy on mobile devices.

B. Transformer for Audio classification

Recently, Transformer architectures [14–20] have become prominent in both visual and auditory recognition tasks. This encoder-decoder framework processes sequential data through self-attention mechanisms. In computer vision, models like ViT [16] segment images into patches, enrich them with positional embeddings, and use transformer encoders for feature extraction before classification via MLP heads. Remarkably, ViT outperforms CNN baselines like ResNet without data augmentation [18], prompting Google's release of large-scale pretrained ViT models.

For audio tasks, the Audio Spectrogram Transformer (AST) [7] adapts ViT to analyze speech spectrograms. Using overlapping patch inputs, AST sets new benchmarks on EST-50, AudioSet, and Speech Commands V2 [21], demonstrating transformers' cross-domain applicability.

C. Knowledge Distillation

Knowledge Distillation (KD) transfers capabilities from large teacher models to compact student networks. Gong et al. propose Cross-Modal KD (CMKD) [22], where a frozen teacher guides a student model bidirectionally—enabling mutual distillation between CNNs and Transformers. Crucially, CMKD proves effective even for efficient models, where distillation further reduces size while maintaining performance. This capability directly motivates our work toward edge-optimized voice control systems.

III. Methodology

In this section, we introduce our dataset and present our approach for constructing Adaptive Window Audio Transformer (AWAT) and distilling it into an extremely-low parameter model. Furthermore, we outline our evaluative method for quantifying the effect of our model and the implementation details of our solution.

A. Adaptive Window Audio Transformer

In this study, an Adaptive Window Audio Transformer (AWAT) is proposed for audio command classification. We utilized an adaptive window Mel-scale Transform for converting varied-length voice signals into same-sized spectrograms, with sizes of 224×224. Spectrograms are then input into audio transformer for classification. And detailed structure of AWAT is shown in Figure. 1.

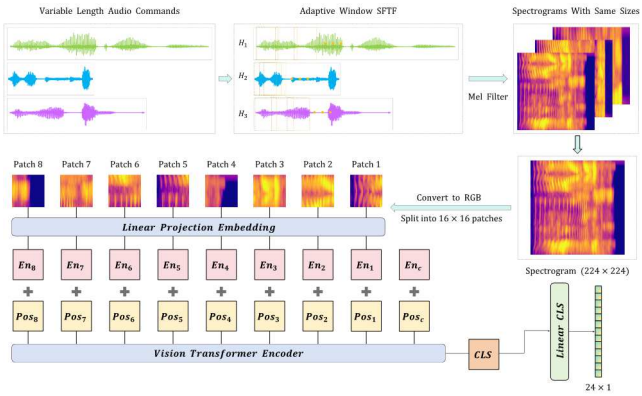


Fig. 1. Detailed structure for AWAT framework.

Unlike traditional Vision Transformer (ViT) architecture based audio classification method, such as AST, AWAT have two advantages:

1. pre-assign input signal length before constructing model is no longer needed for AWAT, which is completely fitted for varied length commands under real scenarios, showing great potential in real application;
2. the size of voice spectrograms is 224×224, which is compatible with the size of input of pre-trained ViT, fully adapting pre-trained weights information and avoiding information loss during resizing operation or overlap patch sampling.

1) Dynamic Hop Length

The Mel-scale transform is a frequency transformation method that mimics the human auditory system's characteristics, widely used in speech signal processing (e.g., speech recognition) and audio feature extraction. Its core concept involves converting linear frequency measurements (Hz) to a nonlinear Mel scale, thereby better aligning with human perception of pitch. In this study, it is utilized in the conversion from 1-D voice signal to 2-D spectrogram.

Given an input audio signal $y(n)$, the silence trimming via $\Delta = 15dB$ thresholding and pre-emphasis filtering $y_{pree}(n) = y(n) - 0.97y(n-1)$ are utilized for signal preprocessing. Thus, the signal is changed into $y(n) \in \mathbb{R}^{L\times 1}$, where L is the length of the signal. Assuming the time frames of target fixed output dimension is $T_{tr} = 224$ and the window length N is set as 4096, the hop length H_{dy} is computed dynamically as:

$$H_{dy} = floor(\frac{L}{T_{tr} - 1}) \quad (1)$$

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where $floor(\cdot)$ denotes the floor division. Then the length of the signal is adjusted to $\bar{L} = H_{dy} \cdot (T_{tr} - 1)$ via zero padding, obtaining $y_{pad} \in \mathbb{R}^{\bar{L} \times 1}$.

2) Power Spectrogram

Fixed Hanning window Short-Time Fourier Transform (STFT) is utilized for spectrogram extraction, which is computed as:

$$\omega(m) = 0.5 - 0.5\cos(\frac{2\pi m}{N}),\ 0 \le m \le N-1 \tag{2}$$

$$X(l,k) = \sum_{m=0}^{N-1} y_{pad}(kS_{dy} + m) \cdot \omega(m) \cdot exp(-j\frac{2\pi lm}{N}) \tag{3}$$

Where $\omega(m)$ is the Hanning window, $0 \le k \le T-1$ indexes time frames and $0 \le l \le N/2$ corresponds to frequency bins.

The power spectrogram is calculated as $P(l,k) = |X(l,k)|^2$ and then mapped to the Mel scale using a filter bank $\{H_q(l)\}_{q=1}^{Q}$, which is constructed through uniform partitioning of the Mel-frequency axis:

$$Mel(f) = 2595\log_{10}(1 + \frac{f}{700}),\quad f = \frac{lF_s}{N} \tag{4}$$

Where $F_s = 22.05kHz$ is the sampling rate and D is the number of filters utilized in this work.

The Mel spectrogram energy is obtained via $E(q,k) = \sum_{l=0}^{N/2} H_q(l)P(l,k)$, followed by logarithmic compression $E_{log}(q,k) = \log_{10}(E(q,k) + 10^{-6})$ and quantile-based normalization:

$$E_{norm}(d,k) = \frac{E_{log}(d,k) - \mu_1}{\mu_{99} - \mu_1} \tag{5}$$

Where μ_p denotes the p-th percentile of E_{log}. Finally, the normalized matrix $M_g = clip(255E_{norm}, 0, 255)$ is gathered, producing a 224×224 grayscale image for further classification.

3) Interface with Pre-trained ViT

Finally, the grayscale image M_g is converted into RGB mode M by repeating triple times. And the chosen structure of ViT is ViT-base[30], which is pre-trained on ImageNet 21k [24] and finetuned on ImageNet 1k [25] by google. The patch size is 16×16 and the input image size is still 224×224. To fully adapt ViT to our work, the final linear layer is replaced with a 24-classification linear layer.

Utilizing an adaptive window for spectrograms construction and adapting ViT into audio transformer, AWAT fully leverages the information that pre-trained weights on ImageNet carried and the Mel spectrograms' capability of aligning human perception of pitch. Consequently, AWAT solves varied-length input signal and information loss issue when adapting pre-trained ViT weights.

B. Knowledge Distillation Framework

Knowledge Distillation transfers expertise from a large teacher model to a compact student network. In real-world edge device utilization scenarios, deploying intelligent models must confront stringent hardware limitations inherent to embedded systems. Edge processors often operate with severely restricted computational throughput, minimal volatile memory allocations, and fixed non-volatile storage capacities, which collectively challenge conventional deep learning architectures.

Consequently, we propose a knowledge distillation framework based on our proposed MiniNet architecture with only 0.26M parameters. This framework bridges the gap between high-accuracy and low-footprint models for edge computing AI applications. Figure. 2 depicts the full knowledge distillation framework in details.

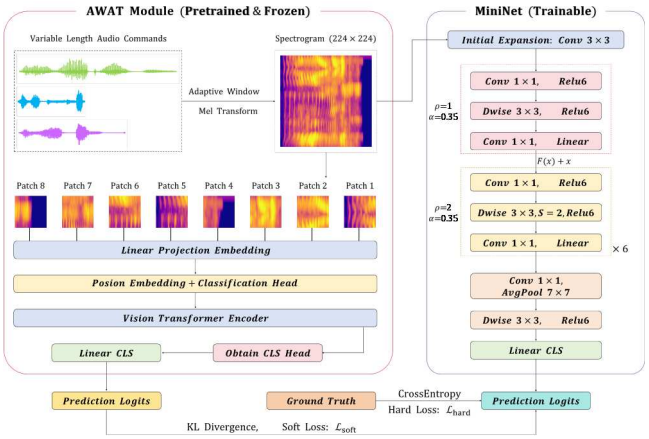


Fig. 2. Detailed process for extreme knowledge distillation.

Our approach achieves two critical advancements:

(1) parameters are substantially reduced: our MiniNet, the student model, inherits the architectural efficiency of MobileNetV2[26] while being further optimized for extreme parameter reduction, enabling direct deployment on resource-limited hardware without external memory dependencies;

(2) task-specific discriminative performance is enhanced: The distilled network demonstrates superior classification accuracy over its teacher model, effectively overcoming the performance saturation bottleneck commonly observed in resource-intensive teacher architectures.

1) MiniNet- The Student Model

MobileNetV2 is a lightweight convolutional neural network architecture designed to mimic the efficient feature selection mechanisms of the human visual system. It is specifically optimized for image classification tasks in resource-constrained environments.

In this study, the student model, i.e., our MiniNet, inherits from MobileNetV2, which is optimized and modified into for mapping Spectrograms to probability distributions across 24 categories.

MiniNet has four components: inverted residual blocks, linear bottlenecks, width scaling and adaptive expansion, which enable efficient nonlinear feature extraction while avoiding redundant computations.

The original MobileNetV2 utilizes inverted residual blocks with linear bottlenecks. And we will formalize its mathematical foundations first, then present our compressed variant. For input three-channel Spectrogram $M \in \mathbb{R}^{H \times W \times C_{in}}$,

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an inverted residual block with expansion ratio t, output channels C_{out} and stride s performs:

$$\bar{M}_{exp} = \mathrm{ReLU6}\left(W_{expand} * \bar{M}\right),\ \bar{M} \in \mathbb{R}^{H\times W\times(t\cdot C_{in})} \tag{6}$$

$$\bar{M}_{dw} = \mathrm{ReLU6}\left(W_{expand} \circledast \bar{M}_{exp}\right)\ \bar{M}_{dw} \in \mathbb{R}^{\hat{H}\times\hat{W}\times(t\cdot C_{in})} \tag{7}$$

$$O = W_{proj} * \bar{M}_{dw},\ O \in \mathbb{R}^{\hat{H}\times\hat{W}\times C_{out}} \tag{8}$$

Eq. (6) is channel expansion operation and usually $t>1$ to increase representational capacity. And Eq. (7) denotes depthwise convolution where $\circledast$ is depthwise operation and $\hat{H}=H/s$, $\hat{W}=W/s$ are spatial downsampling via stride s. Finally, Eq. (8) is simply linear projection, omission of ReLU6 for preventing information destruction in low-dimensional spaces.

2) Dual Compression in MiniNet

We optimize our MiniNet through dual compression, which are width multiplier scaling and adaptive expansion. A global width multiplier $\alpha = 0.35$ scaling all layer channels is introduced as:

$$C_l^{new} = \alpha C_l^{base},\ l \in (1,\cdots,7) \tag{9}$$

Where l is layer index of inverted residual blocks, C_l^{base} is original channel dimension and C_l^{new} is new channel dimension. Meanwhile, adaptive expansion ratio ρ is modified per block to balance capacity:

$$\rho_l = \begin{cases} 1 & l = 1\,(\textit{first block}) \\ 2 & l \geq 2 \end{cases} \tag{10}$$

Finally, MiniNet is constructed as student model, with a total of 0.26M parameters.

3) The Total Loss in Knowledge Distillation

Knowledge Distillation employs soft labels, feature distillation, and adaptive optimization to compress computational complexity while retaining accuracy. In our study, AWAT model is the teacher model and MiniNet is the student model.

Given teacher logits z_t and student logits z_s, the total loss L_{total} of the model composes of soft loss L_{soft} and hard loss L_{hard}.

Soft loss is utilized to make student model mimic the behavior of the teacher model. And the core concept of it is to make a closer logit distribution between z_s and z_t. By utilizing KL-Divergence Loss, student model gradually learns to produce similar logit distribution of teacher, consequently, obtaining vital knowledge from large teacher model and consistently enhancing classification capability. The soft loss is formulated as:

$$\mathrm{L}_{soft} = \sum_{i=1}^{N} \sigma(z_t)_i \log \frac{\sigma(z_t)_i}{\sigma(z_s)_i},\ \sigma(z_k)_i = \frac{exp(z_k^i/\tau)}{exp(\sum_j z_k^j/\tau)} \tag{11}$$

Where N is the total class numbers, which is 24 in this study, and τ is the distillation temperature, which is set to 1.0 in this study.

Hard loss is designed to make student model focus on its classification task directly, which is simply in the form of Cross-Entropy Loss:

$$\mathrm{L}_{hard} = -\sum_{i=1}^{N} y_i \log \sigma(z_s)_i,\quad \sigma(z_s)_i = \frac{exp(z_s^i)}{\sum_j exp(z_s^j)} \tag{12}$$

Finally, total loss is calculated as:

$$\mathrm{L}_{total} = \lambda_1 \mathrm{L}_{soft} + \lambda_2 \mathrm{L}_{hard} \tag{13}$$

Where λ_1 and λ_2 denote weights of soft loss and hard loss, which are set to 1.0 and 0.5, separately.

Combining optimal structure compression and knowledge distillation, our 0.26M MiniNet significantly reduces parameters needed while surpassing classification performance of the teacher model AWAT, which solid demonstrates its great potential in variable length voice identification on edge devices.

IV. Experiment Results

This section presents the experiment dataset, KD implementation and evaluation metrics, ablation and comparative study to validate our model's superiority in addressing the fundamental conflict between computational constraints of edge deployment and real-world acoustic variability. Through our experiments, we demonstrate the framework's operational effectiveness in real-world intelligent voice control applications, confirming its practical viability for resource-constrained embedded systems for edge computing.

A. Dataset

The variable-length audio dataset employed in this study is derived from Intelligent Voice Control directives [23], comprising a set of carefully curated voice commands designed for intelligent voice-controlled systems. The dataset encompasses 24 distinct operational commands including "Bluetooth takes the phone call" and "Bluetooth plays music". The detailed format of voice-controlled directives is provided in Appendix I.

We collected approximately 240 students' voice recordings through mobile devices and laptops, with each participant contributing part of 24 voice command samples. All data were voluntarily provided with explicit privacy consent documentation. The original audio files in diverse formats were filtered and standardized into .wav format through preprocessing. Notably, the temporal duration of voice commands exhibits significant variation, spanning from 1.0 to 8.5 seconds, presenting notable computational challenges in speech processing. The complete dataset containing 5,175 command instances was randomly partitioned into training and testing subsets at an 8:2 ratio.

B. Implementation and Evaluation Metrics

We implement the proposed KD framework using PyTorch 1.13 and Python 3.12, with training and inference conducted on a workstation equipped with eight NVIDIA A800 GPUs. We employed AdamW optimizer for backpropagation, maintaining identical learning rates of 0.0002 for KD-AWAT architecture, with a unified batch size of 16 across all experiments.

The most important goal of the model is to discriminate different audio class labels. Thus, performance is measured as

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prediction accuracy on testing subsets, which consists 1035 audios totally. And the accuracy is formally defined as:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{14}$$

where TP, TN, FP and FN refer to the number of true positive, true negative, false positive, and false negative predictions, respectively.

C. Ablation Study Results

The ablation study evaluates our adaptive window mechanism by comparing four fixed-hop audio transformer (FHAT) configurations (hop lengths: 64, 128, 256, 512) against the proposed AWAT model. During training, the optimizer configurations, such as learning rates, and batch sizes, is set to identical in all models until performance plateaued. Experimental results are presented in Table I.

TABLE I. ABLATION STUDY DETAILS

Method	Time Frame Length			Accuracy
	Average	*Median*	MAE_{224}	
FHAT-64	495	489	270.8	0.9454
FHAT-128	248	245	50.7	0.9513
FHAT-256	124	123	99.6	0.9483
FHAT-512	63	62	161.3	0.9454
AWAT	**224**	**0**	**0**	**0.9600**

Notably, all FHAT configurations maintain a fixed Mel filterbank dimension of 224. This configuration ensures that generated spectrograms exhibit uniform frequency resolution (224 bins) while possessing variable time frame lengths T_i. Specifically, the time frame length T_i dependents on 1) original audio duration, and 2) hop length selection during Mel-scale transform. And finally, for i-th input audio, its spectrogram is obtained in the shape of $224 \times T_i$.

To quantify the deviation between these variable-length spectrograms and the standardized 224×224 format, we compute three statistical measures, including Average, Median and 224-based Mean Absolute Error (MAE224), which are formulated as:

$$Avg = \frac{1}{N}\sum_{i=1}^{N} T_i \tag{15}$$

$$Med = \begin{cases} \hat{T}_{(N+1)/2} & N \text{ is odd} \\ \frac{1}{2}(\hat{T}_{N/2} + \hat{T}_{(N+1)/2}) & N \text{ is even} \end{cases} \tag{16}$$

$$MAE_{224} = \frac{1}{N}\sum_{i=1}^{n} | T_i - 224 | \tag{17}$$

Where N is the total number of spectrograms, and $\hat{T}_i$ is i-th order statistic for a sorted time frame length sequence $(\hat{T}_1, \hat{T}_2, \cdots, \hat{T}_N)$ in ascending order.

As demonstrated in Table II, the proposed adaptive window strategy achieves superior accuracy across all evaluated models. This performance advantage stems from maintaining 224×224 spectrogram dimensions that precisely match the input specifications of pre-trained ViT-base, thereby fully leveraging transfer learning advantages from ImageNet-1k while eliminating feature degradation associated with resolution mismatches.

Comparative analysis of static FHAT configurations identifies FHAT-128 as the optimal variant, with statistical evaluation of temporal metrics (mean, median, and MAE224) confirming its closest alignment to standard frame duration references. This empirical evidence reveals that conventional fixed-window approaches introduce non-linear distortion through improper temporal scaling during spectrogram standardization. Moreover, the performance rank: FHAT-128 > FHAT-256 > FHAT-512 ≈ FHAT-64 and the alignment rank FHAT-128 > FHAT-256 > FHAT-512 > FHAT-64 further demonstrate this conclusion.

The AWAT framework addresses this limitation through dynamic hop length adaptation during time-frequency transform. Through automatically optimizing temporal localization patterns according to input duration characteristics, the system achieves position-wise feature extraction that preserves critical signal relationships. This adaptive processing paradigm circumvents rigid resizing constraints through intelligent temporal resolution regulation, effectively mitigating information loss in variable-length audio inputs.

D. Comparative Study Results

The variable-length audio dataset employed in this study is derived from Intelligent Voice Control directives [23], comprising a set of carefully curated voice commands designed for intelligent voice-controlled systems. The dataset encompasses 24 distinct operational commands including "Bluetooth takes the phone call" and "Bluetooth plays music". The detailed format of voice-controlled directives is provided in TABLE II.

TABLE II. 24 INTELLIGENT VOICE CONTROL DIRECTIVES FOR BLUETOOTH DEVICE

Seq.	Function code	alphabet	The text of voice control directives
0	a	a	Bluetooth device start
1	b	b	Bluetooth device take phone call
2		bb	Bluetooth device phone call
3	c	c	Bluetooth device answer phone call
4		cc	Bluetooth device answer call
5	d	d	Bluetooth device reject phone call
6	e	e	Bluetooth device play music
7		ee	Bluetooth device music
8	f	f	Bluetooth device stop play music
9		ff	Bluetooth device stop music
10	g	g	Bluetooth device last music
11		gg	Bluetooth device last
12	h	h	Bluetooth device next
13		hh	Bluetooth device next music
14	i	i	Bluetooth device increase volume
15		ii	Bluetooth device increase sound
16		iii	Bluetooth device volume up
17		iiii	Bluetooth device sound up
18	j	j	Bluetooth device decrease volume
19		jj	Bluetooth device decrease sound
20	k	k	Bluetooth device shutdown
21	l	l	Bluetooth device battery notice
22		ll	Bluetooth device battery remained
23		lll	Bluetooth device battery

For comparative evaluation, we established a benchmark suite consisting of: 1) Our teacher model AWAT; 2) AST

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variants: AST with adaptive window (AST-AW) and baseline AST; 3) ResNet-18/34/50 [27], VGG-13/16/19 [28], and ConvNeXt-B [29].

Notably, multi-modal approaches combining audio and spectrogram features were excluded due to inconsistent audio durations between training and test datasets. And architecture specifications are listed as follows:

1. **AST-AW:** Implemented adaptive windowing with 128 Mel filters and 100-frame temporal resolution according to the original AST paper.
2. **Baseline AST:** Utilized fixed 256-hop Mel-Scale transform with spectrogram resizing to 128×100 dimensions.
3. **Other models:** Employed adaptive window preprocessing, converting varied length audios into 224×224 spectrogram inputs.

Notably, all compared models followed identical optimization parameters (optimizer, learning rate, batch size) and were trained to convergence. And the final result is shown in Table III.

TABLE III. COMPARISON STUDY DETAILS

Method	FLOPs/G	Parameter	Energy/mJ	Accuracy
AWAT	16.86	85.67M	680.18	0.9600
AST-AW	9.37	86.90M	184.99	0.9094
AST-base	9.37	86.90M	184.99	0.8704
ResNet18	1.82	11.19M	50.33	0.9532
ResNet34	3.68	21.30M	98.11	0.9571
ResNet50	4.13	23.56M	119.96	0.9532
VGG13	11.30	129.05M	153.44	0.8798
VGG16	15.47	134.36M	206.27	0.9049
VGG19	19.63	139.67M	225.80	0.9076
ConvNeXt-B	15.37	87.57M	718.97	0.9516
KD-AWAT	**0.036**	**0.26M**	**12.20**	**0.9660**

The original AST paper specifies that spectrogram input dimensions should be configured as $128\times100t$, where t represents audio duration in seconds. While the authors assert the model's capacity to process variable-length audio inputs, practical implementation reveals substantial dependency on predetermined temporal parameters. Specifically, audio classification applications require explicit duration specification during model initialization. This constraint presents significant operational challenges given the inherent duration variability observed in real-world audio samples, even when containing identical command content. Furthermore, environmental variables and inter-speaker differences exacerbate temporal inconsistencies, collectively diminishing model efficacy and limiting practical deployment potential.

This study configures the input spectrogram dimensions for AST at 128×100 . To evaluate the effectiveness of our adaptive window strategy, we implemented two experimental variants: AST-base and AST-AW. Specifically, the AST-base employs fixed 256-hop Mel-scale transformation for spectrogram conversion, with filterbank size standardized to 128. Resultant spectrograms of varying time frame length dimensions $128\times T_i$ underwent resizing to meet the original AST's 128×100 input specification. The 256-hop configuration was selected based on temporal alignment considerations, as its average (124) and median (123) time frame lengths demonstrate closer proximity to the target time frame length (128) than alternative hop lengths, thereby minimizing information loss during image resizing. And for AST-AW the adaptive window strategy is utilized and the filterbank is set to 128, the target time length T_{tr} is set to 100.

In contrast, AST-AW incorporates our adaptive window strategy while maintaining identical filterbank configuration (128) and target time frame length $T_{tr}=128$. Comparative analysis reveals AST-AW achieves superior prediction accuracy (0.9094 vs. 0.8704) despite equivalent computational complexity (FLOPs), model architecture parameters and average energy consumed per image. This performance enhancement substantiates the adaptive window's capability to maintain precise alignment with pre-trained ViT-base input specifications, thereby optimizing transfer learning benefits from ImageNet-1k while preventing feature degradation caused by resolution mismatches.

Comparative benchmarking demonstrates the proposed KD-AWAT framework achieves optimal predictive accuracy (0.9660) when evaluated against CNN-based architectures and transformer-based approaches. The optimized architecture maintains superior classification performance while exhibiting minimal computational costs (FLOPs), compact efficiency (0.26 million) and energy efficiency (12.20mJ) within the experimental cohort. As visualized in Figures 3-5, these empirical results substantiate that the integration of knowledge distillation with architectural compression enables performance parity surpassing the teacher model while preserving ultra-compact parameterization and energy efficiency.

Thus, our proposed KD-AWAT establishes an optimized equilibrium between computationally intensive architectures and resource-constrained implementations in edge computing environments while preserving excellent prediction performance.

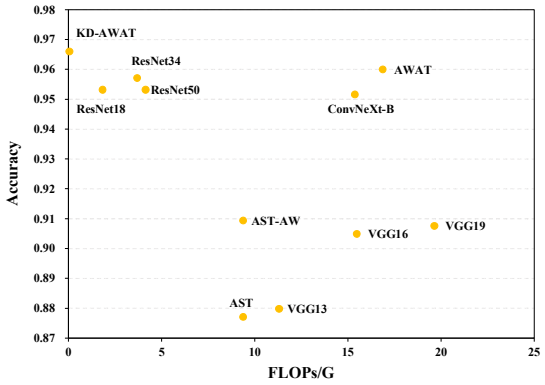


Fig. 3. FLOPs against classification performance for all models.

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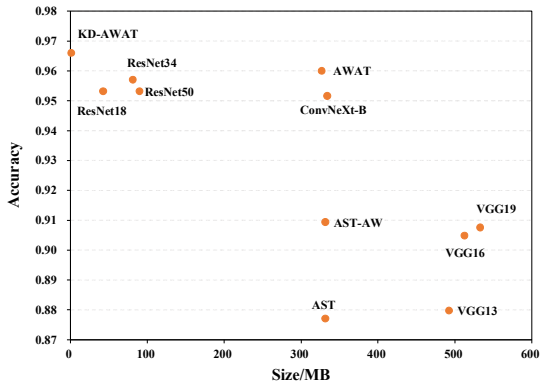


Fig. 4. Classification performance against model size for all models.

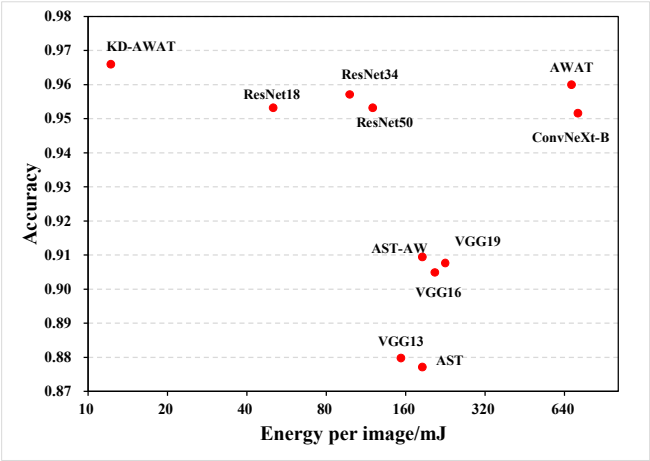


Fig. 5. Classification performance against energy for all models.

E. Experiment Results Summary

The proposed framework was evaluated on a real-world dataset comprising 5,000+ variable-length voice commands (24 command classes, 1.0-8.5s duration) collected from about 240 participants. Comprehensive ablation studies examined AWAT and FLHAT variants with four distinct hop lengths, while comparative analysis included KD-AWAT against teacher AWAT, AST-AW, AST-base, ResNet18/34/50, VGG-13/16/19, and ConvNeXt-B. Key findings reveal:

1. **Adaptive Window Effectiveness:** AWAT achieves 96.0% test accuracy, outperforming all FLHAT configurations with fixed hop lengths, thereby demonstrating the efficacy of adaptive window strategy implementation.
2. **Accuracy-Compression Balance:** KD-AWAT attains 96.6% test accuracy while reducing parameter count by 330× (86M→0.26M) relative to the teacher model AWAT (96.0%), maintaining classification performance, parameter efficiency and energy efficiency superiority over all benchmark architectures.

This empirical evidence confirms KD-AWAT's ability to balance computational demands with operational constraints in edge computing implementations while preserving predictive accuracy.

V. Conclusion and Future Work

Intelligent voice control systems in edge computing require lightweight models capable of processing real-world variable-length audio inputs. While transformer-based models achieve high accuracy, their fixed-length input constraints and excessive parameters hinder deployment in resource-constrained edge environments. This motivates us to construct a framework capable of processing real-world variable length input while maintaining enough accuracy and low parameters. Consequently, KD-AWAT is proposed which utilized adaptive window strategy for varied length audio conversion and structure modification and knowledge distillation for parameter reduction. KD-AWAT bridges the critical gap between transformer-level accuracy and edge-device practicality, while is also capable of dealing with mixed-length audio input, demonstrating significant potential in two key application scenarios:

1. Industrial Automation: Enables real-time voice control systems to process mixed-length audio commands without compromising robustness in dynamic acoustic environments.
2. Energy-Constrained IoT Systems: Achieves efficient deployment on battery-powered edge devices through extreme parameter reduction while retaining high classification accuracy.

Our work validates that adaptive spectrogram processing combined with knowledge distillation provides a viable pathway for deploying variable-length voice control models on resource-limited hardware. Future research will focus on extending KD-AWAT's framework to multilingual support and enhancing robustness against diverse background noise conditions.

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